

# List of major equipment available to carryout Research Activities



राष्ट्रीय प्रौद्योगिकी संस्थान आंध्र प्रदेश

**NATIONAL INSTITUTE OF  
TECHNOLOGY**

**ANDHRA PRADESH**

(An autonomous Institute under the aegis of Ministry of Education, Govt. of India)

Near Kondruprolu, Tadepalligudem, W.G. Dist. – 534101

## Department of Biotechnology

### List of Major Equipment

| S.NO | NAME OF THE INSTRUMENT                            | MAKE & MODEL                     | TECHNICAL SPECIFICATIONS                                                                                                                                                                                                                                                                                                                                                                                                                                                         | FACULTY INCHARGE         |
|------|---------------------------------------------------|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| 1    | Refrigerator<br>Quantity:1                        | Samsung,<br>RR19K173ZR<br>Z/HL   | Type: Single Door<br>Defrosting Type: Direct Cool<br>Compressor Type: Digital Inverter Compressor<br>Capacity:192 L<br>Storage volume: 182L<br>Number of Doors:1<br>Built-in Stabilizer: yes<br>Shelf Material: Toughened Glass Shelves<br>Door Lock: Yes<br>Gasket Type: Anti-Fungal Door<br>Removable Gasket: Yes<br><b>Power Features</b><br>Stabilizer Required: No<br>Number of Refrigerator Shelves:2<br>Number of Refrigerator Drawers:1<br>User Manual and Warranty Card | Dr. T Jagan<br>Mohan Rao |
| 2    | Magnetic Stirrer Hot Plate<br>Quantity:2          | Cole-Parmer<br><br>WW – 15956-32 | OP Plate Material Ceramic-coated Aluminum<br>Min Temperature(°C):20<br>Max Temperature(°C):370<br>No. Of Stirring Positions:1<br>Description: Digital Round Top Ceramic-Coated Stirring Hot Plate with Counter Reaction<br>Stirring speed: 0-1500 RPM                                                                                                                                                                                                                            | Dr. T Jagan<br>Mohan Rao |
| 3    | Refrigerated Circulating Water Bath<br>Quantity:1 | Scientifique<br>IC201-020        | Bath Volume: 20L<br>Temperature Range: -10°C To 100°C<br>Inner Chamber: SS 304<br>Exterior: Powder Coated GI Sheet<br>Lid: SS Lid with Handle<br>Temperature Controller: Digital PID Controller with SV & PV<br>Temperature Sensor:PT100<br>Refrigeration System: CRF Free<br>Heating System: ISI Mark<br>Immersion Heaters<br>Circulation Pump:10 - 15 Liters Per Minute<br>Safety Feature: Over Temperature Cut-Off                                                            | Dr. T Jagan<br>Mohan Rao |

|   |                                           |                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                          |
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|   |                                           |                                                                       | Power Supply:220 Volts 50 Hz                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                          |
| 4 | PH Meter<br>Quantity:2                    | Oakton<br>35413-00                                                    | pH Range 0.00 to 14.00 pH<br>Accuracy $\pm 0.01$ pH<br>Temperature Accuracy: $\pm 0.5^{\circ}\text{C}$<br>Temperature Range:0.0/100.0 $^{\circ}\text{C}$<br>Width :6-7/8”<br>Height :2-3/4”<br>Length :6-1/8”<br>Operating Frequency :60 hertz<br>Specification Met: Certified<br>Frustration-free                                                                                                                                                                                                                                                                                       | Dr. T Jagan<br>Mohan Rao |
| 5 | Electrophoresis Unit<br>Quantity:1        | Hi Eco Mini<br>Horizontal<br>Electrophoresis<br><br>Himedia<br>LA 851 | Robust acrylic frame mounted on non-slip rubber feet.<br>A safety lid fits in one way only preventing access to live components and avoid evaporation. Gel tray gasket provide sealing for leak-proof gel casting parameter. Long life pure Platinum electrodes. Convenient carrying handles enable easy buffer disposal.<br>Operating Temperature range 4-65 $^{\circ}\text{C}$ .<br>Output Range: 50 V / 100 VDC<br>Max. Buffer Volume(ml): 150<br>Gel Casting Tray: 1 No.<br>Hi Eco Mini Horizontal Electrophoresis with maximum samples 14<br>Tank dimensions(L*W*H): 16.2*9.9*2.9cm | Dr. T Jagan<br>Mohan Rao |
| 6 | UV<br>Transilluminator<br>Quantity:1      | Major science,<br>MUV21-312                                           | Dimensions: 21 $\times$ 21 cm<br>Single wavelength 254, 312, or 365 nm.<br>High intensity/low intensity switch for single wavelength models<br>Fast, no blink startup<br>Long lasting filter<br>UV protection cover WITH W*D 330*250 mm<br>Rated voltage :110 V ~/ 220 V~<br>Light source: 8 W *6 Tubes                                                                                                                                                                                                                                                                                  | Dr. T Jagan<br>Mohan Rao |
| 7 | UV-VIS<br>Spectrophotometer<br>Quantity:1 | Shimadzu<br>UV-1800<br>Double beam                                    | Stray Light <0.02%T at 340 nm, 400 nm; <1.0%T at 198 nm<br>Source Lamp Deuterium: Tungsten-Halogen<br>Source Lamp Life :2000 hours<br>Beam Type: Double<br>Bandwidth: 1.0 nm                                                                                                                                                                                                                                                                                                                                                                                                             | Dr. T Jagan<br>Mohan Rao |

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|   |                                          |                                   | Min Wavelength (nm)190<br>Max Wavelength (nm)1100<br>Wavelength Accuracy $\pm 0.1$ nm<br>Wavelength Reproducibility $\pm 0.1$ nm<br>Min Photometric - Transmittance (%T)0<br>Max Photometric - Transmittance (%T)400<br>Min Photometric - Absorbance (A)-4<br>Max Photometric - Absorbance (A)4<br>Photometric Drift<0.0003 A/hr<br>Detector Silicon photodiode<br>Display Type: LCD (4.75" x 3.5")<br>Power: (Hz)60<br>UV/Visible Scanning Spectrophotometer; 115 VA |                       |
| 8 | Vortex Mixer<br>Quantity:2               | Thermo Scientific<br>88880018- LP | Speed: 0 to 3,000rpm<br>Voltage :100 to 240V<br>Hertz:50/60Hz<br>Plug Type: Cord set w/various plugs<br>Type: Mixers<br>Dimensions (L x W x H) Exterior: 8.3 in. x 6.1 in. x 3.3 in., 210 mm x 154 mm x 83 mm<br>Load Bearing Capacity:0.5kg (1.1 lb.)<br>Platform Material: Rubber<br>Operating Modes: Continuous / Touch<br>Electrical Requirements:100 to 240V 50/60Hz                                                                                             | Dr. T Jagan Mohan Rao |
| 9 | Orbital Shaker & Incubator<br>Quantity:1 | Scigenics Biotech,<br>Orbitek LE  | Cabinet Material: Coated Steel (exterior) and Stainless Steel (interior)<br>Current:16 Amp<br>Dimensions External:650 x 750 x 800 mm<br>Dimensions Internal:500 x 500 x 400 mm<br>Phase: Single Phase<br>Platform Size:420 x 420 mm<br>Speed Accuracy: +/- 2% of set value<br>Speed Drive: Brushless AC<br>Temperature Control Accuracy: +/- 0.1 deg C                                                                                                                | Dr. A. Seenivasan     |

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|    |                                        |                                                          | Temperature Uniformity: +/- 0.3 deg C nominal uniformity @ 37.0 deg C in flasks<br>Speed Range:30-400 rpm<br>Frequency:50 Hz<br>Display Type: Digital LED<br>Orbital Diameter:25 mm<br>Temperature Range:5 deg C to 80 deg C<br>Voltage:230 Volt<br>Power:900 W                                                                                                                                                                                                                    |                       |
| 10 | Autoclave<br>Quantity:1                | Equitron, Pad series (# 7421 PAD)                        | Working Chamber Size: 300 * 500 Mm (12x20'') / 35 L<br>Heater: 1.80 Kw<br>Gross Dimensions: 820 X 620 X 1170 Mm Automatic Purging for Efficient Sterilization.<br>Preset (At 121°C) Digital Temperature Indicator Cum Controller Linked to A Preset (20 Min) Timer.<br>Low Water Level Cut Off.<br>Fitted With a Safely Valve / Fusible Safety Plug for Added Safety.<br>End Of Cycle Buzzer.<br>Supplied With Traceable Certificate for Pressure Gauge and Temperature Indicator. | Dr. T Jagan Mohan Rao |
| 11 | Deep freezer<br>Quantity:1             | Blue star 25 Cup right medical freezer (DW – 25 BDL 280) | Capacity(L) :280<br>Temperature Range (°c) -10 To -25<br>Cooling Performance(°c): -25<br>Refrigerant R600a<br>Cooling Method Direct Cooling<br>Display Digital Display<br>Temperature Data<br>Volt/Freq/Power (V/Hz/W) 220-240 /50/135<br>External Material: Pcm<br>Rated Frequency :50 Hz<br>Rated Current :1.8 A<br>Anti- Shock Safety Classification :1<br>Power Consumption: 1.3 Kw / 24 Hr<br>Inner Material Aluminum Plate with Spraying                                     | Dr. T Jagan Mohan Rao |
| 12 | Laminar air flow chamber<br>Quantity:1 | Classic air systems, CASS 900 H                          | Cleanliness: Class 100<br>Particle retention: 0.3 micron<br>Velocity: 90FPM + 20%<br>Illumination: 750 - 800 lumen<br>Noise level: 60-65 decibels                                                                                                                                                                                                                                                                                                                                  | Dr. T Jagan Mohan Rao |

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|    |                                            |                                   | Standard: FED 209E<br>Power supply: 220V single phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                       |
| 13 | Refrigerated centrifuge<br>Quantity:1      | Remi, Neya 16 R (500 - 16000 RPM) | Maximum speed:16000 rpm (fixed angle)<br>Setting RPM: Yes<br>Maximum RCF:21000g<br>Display RCF: Yes<br>Timer: 00:30 to 99:50(mm:ss) and continuous mode<br>Date and Time: Yes<br>Acceleration levels:0-9 (0 = min – 9 = max)<br>Deceleration levels:0-9 (0 = min – 9 = max)<br>Temperature range: –: -10°C to +40°C<br>Precool function: –: Yes<br>Display temperature: –: Yes (°C and °F)<br>Spin function: Yes<br>Programs:10 programs with protection function<br>Indication of rotor: Yes<br>Noise:55 dB:55 dB<br>power supply:220/240 volts, ±10/1Ø,50 HZ AC,15 Amp<br>Rotor1: A 24-2 (max 15000 rpm)<br>Rotor2: A 6-50 (max 9500 rpm)<br>Rotor3: A 6-100 (max 5000 rpm) | Dr. T Jagan Mohan Rao |
| 14 | Microscopes (labomed- LX300)<br>Quantity:1 | Labomed, LX 300 TRINO LED         | Light Source: Halogen or Led<br>Stage Size: 135*125mm, with coaxial drive<br>Illumination: Halogen 6V-20W illumination for LED up to 50000 hours of LED life<br>Voltage: For HL input 220v-240v AC and For LED input 180v-250v with removable imported cord<br>Is It Portable: Yes<br>Halogen Lamp:6V-20W<br>Battery Backup: up to 50000 Hours<br>Viewing Bodies: Binocular or Trinocular tube 45 Degree inclined, 360 Degree rotatable<br>Eyepieces: Focusable eyepiece with graduated scale WF10x/18mm<br>F.O.V. with foldable eye guard, lockable<br>Nosepiece: Quadruple nosepiece (Ball bearing type) with rubber grip.                                                  | Dr.A.Seenivasan       |

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|    |                                        |                      | <p>Par-centered and Parfocal on CNC Machines</p> <p>Objectives: LP series DIN Plan Achromatic objectives 4x, 10x, 40x (spring loaded), 100x Oil (spring loaded)</p> <p>Mechanical Stage: Mechanical stage 135x125mm with co-axial drive, highly smooth on ball slides</p> <p>Input Voltage:220-240 V AC</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                       |
| 15 | SDS-PAGE<br>Quantity:1                 | Bio – Rad            | <p>Casting Stand: Polycarbonate Pin Retaining ring and spring: Stainless steel</p> <p>Casting Frames: Polysulfone</p> <p>Gray gaskets: Thermoplastic rubber (gray)</p> <p>Electrode Assembly: Glass-filled polybutylene terephthalate</p> <p>Electrodes: Platinum wire, 0.010 inches diameter</p> <p>Gasket electrode: inner core Silicone rubber (green)</p> <p>Mini Tank and Lid: Polycarbonate</p> <p>Sample Loading Guides: Delrin</p> <p>Combs: Polycarbonate (10 well) 1 mm</p> <p>Overall Size :(W x L x H, cm) 12 x 16 x 18</p> <p>Precast Gel Compatibility: Ready Gel and Mini-PROTEAN precast gels</p> <p>Voltage Limit 600 V DC and 500 W</p> <p>short plates</p> <p>glass plates with 1.0 mm spacers.</p> <p>power supply:100-120/220-240 volts ,50/60 hz</p> | Dr. T Jagan Mohan Rao |
| 16 | Weighing balance (Macro)<br>Quantity:1 | Ohaus, Scout SPX 421 | <p>Min Capacity :2g</p> <p>Max capacity: 420g</p> <p>Readability :0.1g</p> <p>Repeatability :0.1g</p> <p>Linearity :0.1g</p> <p>Calibration Type: External</p> <p>Output Optional RS232, USB, Ethernet, Bluetooth</p> <p>Stabilization Time :(Sec)1</p> <p>Power: (VAC)115 / 230</p> <p>Depth: (in)8 51/64</p> <p>Platform Width (in)4 45/64</p> <p>Display Column Height (in)25/32</p>                                                                                                                                                                                                                                                                                                                                                                                    | Dr. T Jagan Mohan Rao |

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|    |                                             |                                                        | Description Scout Portable Balance<br>420G x 0.1g                                                                                                                                                                                                                                                                                                                                                                                                                                               |                          |
| 17 | Weighing balance<br>(Micro)<br>Quantity:1   | Afcoset<br>ER120A                                      | Min Capacity: 100 mg<br>Max Capacity: 220gm<br>Power supply: 220/240 v AC (50<br>HZ)<br>Pan diameter :85mm/3.3”<br>Stabilization time: approx. 5 sec                                                                                                                                                                                                                                                                                                                                            |                          |
| 18 | Weighing balance<br>(sunrise)<br>Quantity:1 | Sunrise<br>Technology,<br>FB224                        | Type of Weighing Scale: Digital<br>Capacity:220GM X 0.1MG Scale<br>Calibration: Internal<br>Accuracy :0.1 mg<br>Min :0.1 mg<br>Max Capacity: 220gm                                                                                                                                                                                                                                                                                                                                              | Dr. T Jagan<br>Mohan Rao |
| 19 | Ice Flaking Machine<br>Quantity:1           | BR<br>BIOCHEM<br>Life sciences<br>pvt ltd, BRIF-<br>50 | Ice making capacity :50 kg/24 hr<br>Ice storage :15 kg<br>Voltage:220v/50 Hz<br>Power supply :380 W<br>Cooling type: Air cooling<br>Refrigerant: R134a<br>Water supply: tap water<br>Dimension(mm):400*543*720<br>W*D*H (INCH):18.4*24.2*30.7                                                                                                                                                                                                                                                   | Dr. T Jagan<br>Mohan Rao |
| 20 | Water Bath (Mini)<br>Quantity:1             | Grant JBN5                                             | Bath volume:5L<br>Temp range: Ambient +5 to 95°C<br>Stability (DIN 12876) @70°C: 0.5°C<br>Temperature setting: Digital<br>Supply voltage: 120 or 230V                                                                                                                                                                                                                                                                                                                                           | Dr. T Jagan<br>Mohan Rao |
| 21 | Autoclave (Mini)<br>Quantity:1              | Gera Ventures<br>NIS-115A                              | Capacity: 30L<br>Supply voltage: 120 or 230V<br>Operating system: manual<br>Supplied With Pressure Gauge and<br>Temperature Indicator                                                                                                                                                                                                                                                                                                                                                           | Dr. T Jagan<br>Mohan Rao |
| 22 | Magnetic Heater<br>Stirrer<br>Quantity:2    | Lab Man<br>Scientific<br>Instruments,<br>mLabs-5 LC    | Stirring speed: 100-1600RPM<br>Temp range: ambient-100°C<br>Size: Dimension LxWxH (mm) 300<br>x 190 x 126<br>Display: Digital LCD display with<br>back light<br>Big Size Ceramic hot plate with<br>flame protection and durability<br>Heating and stirring can proceed<br>simultaneously<br>Temperature & Stirring speed is<br>stepless adjusted.<br>Real temperature & speed display<br>on the screen simultaneously<br>Two sensors, outside PT 100 sensor<br>controls the liquid temperature, | Dr. T Jagan<br>Mohan Rao |

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|    |                                           |                           | inside K type sensor control the hot plate temperature (selectable)<br>Over temperature alarm<br>Standard accessories: Stir Bar, Temperature Probe, Stand, Instruction Manual.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                       |
| 23 | Leica Dark Field Microscope<br>Quantity:1 | Leica Microsystems DM 500 | Head: 45° binocular tube<br>Eyepiece 10X/20 eyepiece w/eye guard<br>Objective: maximum of 10x (0.25 NA), 20x (0.4 NA), 40x (0.65 NA), & 100x/Oil (NA 1.25) Objective lens: Abbe Condenser NA at Dry 0.9 and for oil 1.25<br>Phase Slider: 4 positions<br>Bright field: 10X/20X/40X/100x<br>Features: Should have Co-axial focusing, Micrometer arrangement: Fine and coarse adjustment<br>Camera: Minimum 5-megapixel cameras that can live-stream HD images to students' smartphones or tablets via Wi-Fi.<br>Software is free to download and work with android and windows-based system<br>The camera can also be accessed with any Wi-Fi enabled Windows PC or MAC<br>For standalone HD screens or larger screen projections you can use the HDMI connection<br>Lamp: LED with minimum 6000 K temp, 25,000 h life at full intensity<br>Warranty: 5 Year Mechanical and 1 year electrical<br>Accessories: Immersion oil, Dust cover, power cord, Manual, phase contrast slider & camera module<br>Binocular bright, dark fields & phase contrast | Dr.A.Seenivasan       |
| 24 | LG Microwave Oven<br>Quantity:1           | LG, MC3286BLT             | Capacity: 32L<br>Convection: can be used for baking along with grilling, reheating, defrosting and cooking<br>Input power: 230V~50H<br>Frequency: 2450MHz<br>Includes: shelf, glass tray and tawa and rotating ring                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Dr. T Jagan Mohan Rao |

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| 25 | Hot Air Oven<br>Quantity:1                | Equitron,<br>Oven-Stream<br>Series (#7051-<br>250L) | Capacity: 250L<br>Temp range: +50°C to 200°C.<br>Control accuracy: ±0.5°C.<br>Uniformity: ±2.0 at 100°C<br>Inner dimensions(W*D*H):<br>600*600*700mm<br>Outer dimensions(W*D*H):<br>790*930*1050mm<br>Construction: external structure of<br>steel, duly epoxy coated, rest SS304<br>Shelves: 3 stainless steel wire mesh<br>shelves. Shelf height adjustable in<br>25mm steps.<br>Glass window in-built into the door.<br>Automatic cut off of heater &<br>blower when door opened.<br>Digital PID temperature controller<br>with timer, alarms and auto tuning.<br>Non-contact type door switch.<br>Aero dynamic internal design for<br>achieving horizontal air circulation.<br>Electric supply: 230V AC, 50Hz,<br>Single phase.<br>Gross weight: 145kg.<br>CE certified. | Dr. T Jagan<br>Mohan Rao |
| 26 | Double Distillation<br>Unit<br>Quantity:1 | Borosil,<br>3365041                                 | Material: All quartz<br>Distillate temp:65-75°C<br>Dist. Water output<br>capacity(approx.): 1.5l/hr<br>Electrical requirements:230-250V<br>single phase 3kw<br>Minimum cooling water<br>requirement: 1l/hr<br>Biological activity: pyrogen free<br>Conductivity s/cm: <1x10 <sup>-6</sup><br>Total organic carbon (TOC) µg/l<br>:<500<br>Organic matter mg/l: nil<br>Total solids mg/l: <0.1<br>Total plate count: 0<br>Standard plate count: 0<br>Yeast and mold: 0<br>KmnO <sub>4</sub> color retention: 1hr<br>UV absorbance at 254nm: 0.007                                                                                                                                                                                                                              | Dr.A.Seenivasan          |
| 27 | Whirlpool<br>Refrigerator<br>Quantity:1   | Whirlpool,<br>IF515D<br>500Ltrs                     | Energy Efficiency:3 Star Rating<br>Capacity:500 liters<br>Annual Energy Consumption:248<br>Kilowatt Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Dr. T Jagan<br>Mohan Rao |

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|    |                         |                           | Refrigerator Fresh Food<br>Capacity:382 liters<br>Freezer Capacity:113 Liters<br>Form Factor: Standard double door<br>Special Features: Inverter,<br>Convertible, Toughened Glass,<br>Stabilizer Free Operation, Door<br>Lock<br>Cooler: Black<br>Voltage:230 Volts<br>Number of Drawers:2<br>Defrost System: Frost Free<br>Door Orientation: Right<br>Shelf Type: Glass<br>Number of Shelves:3<br>Certification: Energy Star<br>Material: Steel<br>Included Components: Refrigerator,<br>User manual, Warranty card                                                                                                                                                                                                                             |                 |
| 28 | Fermenter<br>Quantity:1 | Eppendorf,<br>BIO FLO 120 | <b>Fermenter:</b><br>working volume of 1 to 2 L (Total<br>capacity of vessel - 3 L)<br>Controllers capable of handling two<br>different platforms i.e., glass and<br>single use vessels with volumes<br>ranging from 250 mL to 40 L on<br>interchangeable basis.<br>Suitable both microbial<br>fermentation as well as Mammalian<br>cell culture<br>Fermenter is Autoclavable<br>Temperature control: 20°C to<br>maximum 60°C<br>Stirrer motor for microbial<br>fermentation operational from a<br>minimum of 100 rpm to 1200 rpm.<br><b>Chiller:</b><br>Recirculating Chiller suitable for<br>the operation of 3 L fermenter.<br>Capacity: 7 liters or above<br>Temperature range: -10 to 100 °C<br>Digital temperature controller<br>available | Dr.A.Seenivasan |

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|  |  |  | <p>The material of construction is resistant to corrosion</p> <p>It provides provision for drain</p> <p>Auto restart and auto cut off option available.</p> <p><b>Air compressor</b></p> <p>Air Compressor compatible with 3 L fermenter</p> <p>Compressor is light weight, compact, oil free and rocking piston type. Air vents available</p> <p>Motor H.P: 0.75 – 1, Noise level (dB): 60 or less, Max. Pressure bar: 7</p> <p><b>Autoclave</b></p> <p>Autoclave suitable for sterilization of a 3 L fermenter.</p> <p>Autoclave have internal basket which can hold fermenter with all components for sterilization purpose.</p> <p>Material of construction: double walled stainless steel</p> <p>Autoclave dimensions – internal: min. diameter - 450 mm and min. height - 700 mm.</p> <p>Nitrogen cylinder included.</p> |  |
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## DEPARTMENT OF CIVIL ENGINEERING

### List of Major equipment

| Concrete Technology Lab      |                                                                                                                                      |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| S.NO                         | Name of the equipment /Item                                                                                                          |
| 1                            | Vibrating table IS:2514-1963                                                                                                         |
| 2                            | Hot air oven                                                                                                                         |
| 3                            | Needle vibrator                                                                                                                      |
| 4                            | 4 Piller Compression Testing Machine                                                                                                 |
| 5                            | Vee Bee Consistometer as per IS: 10510-1983                                                                                          |
| 6                            | Vibrating Machine as per IS :10080-1982                                                                                              |
| 7                            | Rebound Hammer test instrument with all accessories                                                                                  |
| 8                            | Concrete drum mixer                                                                                                                  |
| 9                            | Electronic weighing balance 100KG readability 10gm                                                                                   |
| 10                           | Electromagnetic sieve shaker                                                                                                         |
| 11                           | Flow table( electrically operated)                                                                                                   |
| 12                           | Concrete Permeabilty Apparatus 3 cell model for 100 mm cube                                                                          |
| 13                           | Concrete Permeabilty Apparatus 3 cell model for 150 mm cube                                                                          |
| 14                           | Accelerated Curing Tank                                                                                                              |
| 15                           | Bouyance Balance Apparatus                                                                                                           |
| Geotechnical Engineering Lab |                                                                                                                                      |
| S.NO                         | Name of the equipment /Item                                                                                                          |
| 1                            | HS31.515 Automatic Sieve Shaker                                                                                                      |
| 2                            | HOS Thermostatically Controlled Hot Air Oven                                                                                         |
| 3                            | HS 12.10 Permeability Apparatus (For determenation of coffcient of permeability by constant head parameter & variable head parameter |
| 4                            | HS 22.15 Unconfined Compression Apparatus                                                                                            |
| 5                            | HS 24.15 Direct shear apparatus, load frame 2 KN Capacity with proving ring and dail gauges                                          |
| 6                            | HS 24.617: Electronic Direct Shear Apparatus; 12 speed Motorised with Microprocessor based touch panal electronic unit               |
| 7                            | HS 18.35 : Three Gang Bench Type Consolidation Apparatus                                                                             |
| 8                            | Hydrometer analysis Set up ( Soil hydrometer, motor operated stirrer, digital thermometer , sodium hexametaphosphate)                |
| 9                            | HS 16.07: Soil sample extruder with hydraulic jack system                                                                            |
| 10                           | Motor Driven Automatic Soil Compactor                                                                                                |
| 11                           | Triaxial shear test and unconfined compressive strenght test apparatus : Elettrically operated                                       |
| Surveying Lab                |                                                                                                                                      |
| S.NO                         | Name of the equipment /Item                                                                                                          |
| 1                            | Hand Held GPS                                                                                                                        |
| 2                            | California Bearing Ratio test Apparatus                                                                                              |
| 3                            | Electronic Digital Theodolite 1/2 accaracy                                                                                           |
| 4                            | Electronic Total Station (Refractor less)                                                                                            |

| Environmental Engineering Lab |                                                               |
|-------------------------------|---------------------------------------------------------------|
| S.NO                          | Name of the equipment /Item                                   |
| 1                             | PH Meter                                                      |
| 2                             | Handheld Conductivity Meter                                   |
| 3                             | Benchtop Dissolved Oxygen Menter                              |
| 4                             | Handy Dissolved Oxygen Menter                                 |
| 5                             | Turbidity Meter                                               |
| 6                             | Micro Centrifuge                                              |
| 7                             | Oven                                                          |
| 8                             | Vacuum Pump                                                   |
| 9                             | High Temperature Horizontal Muffle Furnace                    |
| 10                            | Weighing Balance                                              |
| 11                            | BOD Incubator                                                 |
| 12                            | Refrigerator                                                  |
| 13                            | COD Digester                                                  |
| 14                            | Jar Test Apparatus                                            |
| 15                            | Research Centrifuge                                           |
| 16                            | Orbital Shaking Incubator                                     |
| 17                            | Mini Rotary Shaker                                            |
| 18                            | UV-Visible Spectrometer                                       |
| 19                            | Ultrapure RO+MilliQ                                           |
| 20                            | Autoclave                                                     |
| 21                            | Laminar Hood                                                  |
| 22                            | Ion Selective Electrode (ion meter, fluoride,bromide& iodide) |
| FM & HM LAB                   |                                                               |
| 1                             | Reynold's Apparatus                                           |
| 2                             | Pitot Static Tube Apparatus                                   |
| 3                             | Pipe Friction Apparatus                                       |
| 4                             | Open Circuit SubSonic Tunnel                                  |
| 5                             | Metacentric Height Apparatus                                  |
| 6                             | Bernoullis Theorem Apparatus                                  |
| 7                             | Tilting Fume                                                  |
| 8                             | Centrifugal Pump Test Rig                                     |
| 9                             | Francis Turbine                                               |
| 10                            | Kaplan Turbine                                                |
| 11                            | Pelton Wheel Turbine                                          |
| Engineering Geology Lab       |                                                               |
| S.NO                          | Name of the equipment /Item                                   |
| 1                             | Spectrometer                                                  |
| 2                             | Laboratory Polarising Microscope "Radical" Make               |

## Department of Chemical Engineering

### List of Major equipment

| S. No. | Name of the Equipment                                                    | Model Number | Technical Specification                                                                                                                                                                                                                                                                                                                                                                                           | Faculty-in-charge     |
|--------|--------------------------------------------------------------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| 1      | Iso Thermal Batch Reactor                                                | CLCRE 07     | Temp. Sensor : RTD PT-100 type<br>Control panel comprises of :<br>Digital Temp.PID Controller : 0-199.9°C, (For Water Bath)                                                                                                                                                                                                                                                                                       | Dr. Vinoth Kumar Raja |
| 2      | Plug Flow Tubular Reactor(Coiled tube type & Compressed air feed system) | CLCRE 05     | Reactor : Material Stainless Steel, Volume (0.6-0.7) Ltrs (approx.). Feed Tank (2Nos.) : Material Stainless Steel, Capacity - 20 Ltrs. Pressure Regulator : 0-2 Kg/cm <sup>2</sup> . Pressure Gauge : Bourdon type 0-2 Kg/cm <sup>2</sup> .                                                                                                                                                                       |                       |
| 3      | Continuous Stirred Tank Reactor(CSTR)                                    | CLCRE01      | Pressure Regulator : 0-2 Kg/cm <sup>2</sup><br>Pressure Gauge : Bourdon type 0-2 Kg/cm <sup>2</sup> (ISI Make)                                                                                                                                                                                                                                                                                                    |                       |
| 4      | Combined Flow Reactor(Compressed air feed system)                        | CLCRE 11     | Pressure Regulator : 0-2 Kg/cm <sup>2</sup><br>Pressure Gauge : Bourdon type 0-2 Kg/cm <sup>2</sup>                                                                                                                                                                                                                                                                                                               |                       |
| 5      | Cascade Continuously Stirred Tank Reactor(Compressed air feed system)    | CLCRE 03     | Pressure Regulator : 0-2 Kg/cm <sup>2</sup><br>Pressure Gauge : Bourdon type 0-2 Kg/cm <sup>2</sup> (ISI Make)                                                                                                                                                                                                                                                                                                    |                       |
| 6      | Packed Bed Reactor(Compressed air feed system)                           | CLCRE 10     | Reactor Column : Material Borosilicate Glass, Volume 1.2 Ltrs. (approx). Packing : Rasching Rings, Material Borosilicate Glass. Size 6-8 mm (approx.). Feed Tank (2Nos.) : Material Stainless Steel, Capacity 20 Ltrs. Feed Circulation : By Peristaltic Pump (2Nos.). Computer controlled for different RPM. Piping : Stainless Steel and Silicon pipe. Product Analysis : By Electronic Sensor, Output 4-20 mA. |                       |

|                          |                                                                         |                 |                                                                                                                                                                                                                                                                                              |                          |
|--------------------------|-------------------------------------------------------------------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| 7                        | Condensation Polymerization Set-Up                                      | CLCRE 19        | Condensate collecting tank :<br>Material Stainless Steel,<br>Capacity 5 Ltrs. Temperature<br>sensors : RTD PT-100 type<br>Control panel comprising of :<br>Digital Temperature Controller :<br>0-199.9°C RTD PT-100 type<br>(For Reactor).<br>RPM Indicator : Digital, Non-<br>contact type. |                          |
| 8                        | Fixed Bed Catalytic Reactor                                             | CLCRE 26        | The feed flow rate variable<br>between 0 and 20ml/min.<br>The FIA pump able to give flow<br>rates up to 2.5ml/min.                                                                                                                                                                           |                          |
| 9                        | RTD Studies in CSTR                                                     | CLCRE 14        | Pressure Regulator : 0-2 Kg/cm2<br>Pressure Gauge : Bourdon<br>type 0-2 Kg/cm2                                                                                                                                                                                                               |                          |
| 10                       | RTD Studies in PFR                                                      | CLCRE 15        | Pressure Regulator : 0-2<br>Kg/cm2.<br>Pressure Gauge : Bourdon type<br>0-2 Kg/cm2.                                                                                                                                                                                                          |                          |
| 11                       | Chemical Reactor Trainer                                                | CLCRE 16        | Measurement of conductivity =<br>0-200 mS<br>Temperature = 0-1000C<br>Tubular reactor: A 20-30 meter                                                                                                                                                                                         |                          |
| MASS TRANSFER LABORATORY |                                                                         |                 |                                                                                                                                                                                                                                                                                              |                          |
| S.<br>No.                | Name of the Equipment                                                   | Model<br>Number | Technical Specification                                                                                                                                                                                                                                                                      | Faculty-in-<br>charge    |
| 1                        | Liquid-Liquid Extraction in<br>Packed Bed                               | KCMT 101        | Diameter of column:0.048m,<br>column height:0.75m                                                                                                                                                                                                                                            | Dr. Vinoth<br>Kumar Raja |
| 3                        | Sieve Plate Distillation<br>Column                                      | KCMT 112        | No. of plates:07                                                                                                                                                                                                                                                                             |                          |
| 4                        | Packed Bed Distillation<br>Column                                       | KCMT 113        | Packed height:0.90m                                                                                                                                                                                                                                                                          |                          |
| 6                        | Absorption in Packed Bed                                                | KCMT 122        | Diameter of the<br>column:0.048m,column<br>height:0.750m                                                                                                                                                                                                                                     |                          |
| 8                        | Solid In Air Diffusion<br>Apparatus                                     | KCMT 132        | Diameter of column:0.050m                                                                                                                                                                                                                                                                    |                          |
| 9                        | Forced Draft Tray Dryer                                                 | KCMT 142        | Diameter of orifice:0.026m,<br>diameter of pipe:0.052m                                                                                                                                                                                                                                       |                          |
| 10                       | Mass Transfer with/without<br>Chemical Reacion(Solid-<br>Liquid System) | KCMT 144        | Rod diameter:0.007m                                                                                                                                                                                                                                                                          |                          |
| 11                       | Adsorption Studies                                                      | KCMT 165        | Length of column:0.5mtrs,<br>diameter of column:70mm                                                                                                                                                                                                                                         |                          |
| 12                       | Liquid Liquid Extraction                                                | KCMT<br>102A    | Diameter of column:0.048m,<br>column height:1m                                                                                                                                                                                                                                               |                          |

| PROCESS CONTROL AND INSTRUMENTATION LABORATORY |                                   |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                            |
|------------------------------------------------|-----------------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| S. No.                                         | Name of the Equipment             | Model Number | Technical Specification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Faculty-in-charge          |
| 1                                              | Flapper Nozzle System             | BSP20        | Flapper Nozzle System : Combined With Flapper & Nozzle assembly. Micrometer : For Linear displacement measurement. Pressure Regulator : Compatible range. Pressure Gauge : Compatible range. Piping : Size ¼”.                                                                                                                                                                                                                                                                                                                    | Dr. P. Dinesh Sankar Reddy |
| 3                                              | Control Valve Characteristics     | BSP08        | Control Valve : 2Nos. Characteristics : Linear & Equal % . Type : Pneumatic, Size : ½”. Actuator : 15 Sq. Inch. Stroke : 14 mm. Input : 3-15 PSIG. Water Tank : Material Stainless Steel, Capacity 25 litres. Water Circulation : FHP Pump Champion/Standard make. Overhead Tank : Material Stainless Steel, Capacity 10 Ltrs. Flow Measurement : By Rotameter. Pressure Head measurement : By Single column manometer. Pressure Regulator : 0-2 kg/cm². Pressure Gauge : Bourdon type, 0-2 kg/cm². Piping : Size ½”              |                            |
| 4                                              | Characteristics of PID Controller | BSP01Y       | Temperature:40-640 °C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                            |
| 5                                              | Temperature Control Trainer       | BSP17Y       | Temperature Transmitter : Input RTD PT-100 (Range 0-100°C), Output 4-20 mA. Process tank : Material Stainless Steel, Capacity 0.5 lit (approx.). Heater : Nichrome Wire Heater, Capacity 1 kW Thyristor Controller : Input 4-20mA for heater. Flow Measurement : By Rotameter. Piping : Size 1/4”. Interfacing unit : For input-output communication with auto/manual facility. Micro-processor Controller : PID Setting, auto tuning, fully programmable with serial. communication Software : For experimentation, PID control, |                            |

|   |                          |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
|---|--------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|   |                          |           | Data logging, trend plot, offline analysis and printing.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |
| 6 | Flow Control Trainer     | BSP05-D   | Rotameter range:245-2450 LPH, head type flow transmitter output: 4-20mADC, HART                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
| 7 | Level Control Trainer    | BSP06-D   | <p>Level Transmitter : Range 0-300 mm, Capacitance Type, Output 4-20 mA. Process tank : Material SS with Scale. Capacity 2 Ltrs approx. Water Tank : Material Stainless Steel, Capacity 10 Ltrs. Water Circulation : FHP Pump. Champion/Standard make. Flow Measurement : By Rotameter. Control valve : Compatible capacity with pneumatic actuator. I/P converter : Input 4-20mA, Output 3-15 PSIG.</p> <ul style="list-style-type: none"> <li>• Pressure Regulator : 0-2 kg/cm<sup>2</sup>.</li> <li>• Pressure Gauge : Bourdon type, 0-2 kg/cm<sup>2</sup></li> <li>• Piping : Size ¼"</li> <li>• Interfacing unit : For input-output communication with auto/manual facility</li> <li>• Micro-processor Controller : PID Setting, auto tuning, fully programmable with serial Communication</li> <li>• Software : For experimentation, PID control, Data logging, trend plot, offline analysis and printing.</li> </ul> |  |
| 8 | Pressure Control Trainer | BSP07-2-D | <p>Pressure Transmitter : Range 0-5 bar, type strain gauge, output 4-20 mA.</p> <ul style="list-style-type: none"> <li>• Process Tank : Material Stainless Steel, Capacity 1.5 Ltrs.</li> <li>• Control Valve : Compatible capacity with Pneumatic</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |

|                                          |                                      |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |
|------------------------------------------|--------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|                                          |                                      |        | <p>Actuator.</p> <ul style="list-style-type: none"> <li>• I/P converter : Input 4-20mA, Output 3-15 PSIG.</li> <li>• Pressure Regulator : 0-2 kg/cm<sup>2</sup>.</li> <li>• Pressure Gauge : Bourdon type, 0-2 kg/cm<sup>2</sup>, 0-7 kg/cm<sup>2</sup></li> <li>• Piping : P.U.</li> <li>• Interfacing unit : For input-output communication with auto/manual facility</li> <li>• Micro-processor Controller : PID Setting, auto tuning, fully programmable with serial communication</li> <li>• Software : For experimentation, PID control, Data logging, trend plot, offline analysis and printing</li> </ul>                                                                                                                                                                                              |  |
| 9                                        | Interacting & Non-Interacting System | BSP06Y | <p>Process Tank : Material Stainless Steel, Circular, with graduated level scale (3 Nos.) Capacity 3.5 litres each (approx.)</p> <ul style="list-style-type: none"> <li>• Supply Tank : Material Stainless steel, Capacity 20 litres.</li> <li>• Overhead tank : Material Stainless steel, Capacity 5 litres.</li> <li>• Water Circulation : FHP Pump, Champion/Standard make.</li> <li>• Piping : SS &amp; PVC, size ¼"</li> <li>• Flow Measurement : By Rotameter.</li> <li>• The whole unit is assembled rigidly on a base plate.</li> <li>• An ENGLISH instruction manual consisting of experimental procedures, block diagram etc. will be provided along with the Apparatus.</li> <li>• The whole set-up is well designed and arranged on a rigid structure painted with industrial PU Paint.</li> </ul> |  |
| CHEMICAL ENGINEERING RESEARCH LABORATORY |                                      |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |

| S. No. | Name of the Equipment           | Model Number     | Technical Specification                                        | Faculty-in-charge                                 |
|--------|---------------------------------|------------------|----------------------------------------------------------------|---------------------------------------------------|
| 1      | Hydraulic Press                 | Velan Eng.       | 1 ton capacity                                                 | Dr. P. Dinesh Sankar Reddy & Dr.Vinoth Kumar Raja |
| 2      | Mechanized Extruder             | VBCC             | 5.5 inner diameter, 11.5 outer diameter                        |                                                   |
| 3      | High Temperature Muffle Furnace | BST/MF/1 4508812 | Working Temp:1350 <sup>o</sup> C Max. temp:1450 <sup>o</sup> C |                                                   |
| 4      | Hot Air Oven                    | BST/HAO-1123     | Volume: 45 ltrs                                                |                                                   |
| 5      | Ultra Sound Sonicator           | USB 3.5 H/DTR    | Capacity 3.5 ltrs                                              |                                                   |
| 6      | Laboratory Refrigerator         | RLR-300          | Temperature:1-10 <sup>o</sup> C                                |                                                   |
| 7      | Electronic Weighing Scale       | MAB 220          | 220gm*0.1mg                                                    |                                                   |

## DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

### R&D facilities: List of Major equipment

| -S. No. | Name of the Equipment      | Model Number             | Technical Specification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Faculty-in-charge     |
|---------|----------------------------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| 1.      | Cluster of 5 Rack Servers  | HPE ProLiant ML350 GEN10 | <p>Processor: 2 * Intel Xeon Bronze 3104<br/>1.7GHz/9.6GT/2133MHz/8.25MB/6C/Non-HT/U0/85W</p> <p>Memory: 2 * DDR4 2666 RDIMM 16GB</p> <p>Network: 1 * Dual Port Intel i350-AM2 Gigabit LAN controller + 1*Mgmt LAN</p> <p>Storage Bay: 8 * Hot-Swap 3.5/2.5" HDD Trays</p> <p>Expansion Slots: Full-length/Full-height<br/>8* PCI-E 3.0 * 16(4 at * 16 Link or 8 at x8 Link)<br/>Half-Length/Low-Profile</p> <p>Rear:<br/>1*PCI-E 3.0*24(support with riser card)<br/>(1*PCI-E*16(*16 Gen3 Link) and 1*PCI-E *8 (*16Gen3 Link))</p> <p>Front:<br/>1*PCI-E *8(internal HBA/RAID card)</p> <p>Graphics: 2 * Quadro P2000 5GB</p> <p>Storage: 3 * 600 GB SAS HDDs (or) 3.5" SATA3 HDD 2TB 7200RPM</p> <p>Raid: Intel RAID supporting 0/1/5</p> <p>Front I/O Ports: 2 * USB 3.0 Ports + 2 * USB 2.0 Ports</p> <p>Rear I/O Ports: 2 * USB 3.0 Ports,1*VGA port, 1 * RJ-45 Mgmt LAN port,2 *RJ-45 GbE LAN ports</p> <p>Power Supply: 2 * 1600W Power Supply</p> | Dr. Karthick Seshadri |
| 2.      | 64-Core Multi-core Machine | Intel Xeon 7210          | <p>Processor: Intel Xeon 7210, up to 1.50 GHZ, 64 core, 256 Threads</p> <p>Mother Board: Intel S7200 APR family</p> <p>Memory: 64 GB DDR4 LRDIMM</p> <p>Storage: 500 GB SSD and 1 x 2 TB HDDS</p> <p>Cache: 32 MB on-CPU Cache</p> <p>Chipset: Intel C-612 chipset</p> <p>Expansion: 1 x PCIe 3.0 x 16 or 1 x PCIe Gen3 X 20 (x16 or x4)</p> <p>Ethernet: 2 x ethernet Ports</p> <p>External IO: 2 x USB 3.0</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Dr. Karthick Seshadri |
| 3.      | Tower Server               | Dell Precision 5820      | <p>Processor: Intel Xeon W-2265 CPU 3.50GHz, 3504 Mhz, 12 Cores, 24 Logical Processors</p> <p>Mother Board: Intel Xeon Processors P family/Core i7 IOxAPIC</p> <p>Memory: 32 GB DDR4</p> <p>Storage: 500 GB HDD</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Dr. Karthick Seshadri |

|    |                            |                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                       |
|----|----------------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
|    |                            |                           | <p>Cache: 32 MB on CPU Cache</p> <p>Chipset: Intel 200 Series Chipset Family LPC Controller (C422)-A2D3</p> <p>Ethernet: 2x ethernet ports</p> <p>External I/O: 8 USB 3.0</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                       |
| 4. | Tower GPU Server           | Optiplex 7090             | <p>Processor: 10th Generation Intel Core i7-10700 (Octal core, 16 MB Cache, 2.9 GHz to 4.8 GHz, 65 W)</p> <p>Graphics Card: NVidia Quadro T600 8GB GDDR6</p> <p>Memory: 16 GB, 2 x 8 GB, DDR4 non ECC memory</p> <p>Storage: 1 TB 7200 RPM SATA HDD</p> <p>1 x Ethernet port.</p>                                                                                                                                                                                                                                                                                                                                                                               | Dr. Karthick Seshadri |
| 5. | 64-Core Multi-core Machine | Intel Xeon 7210           | <p>Processor: Intel® Xeon 7210, upto 1.50 GHz, 64 core, 256 Threads</p> <p>L2 Cache: 32MB on-CPU Cache</p> <p>Chipset: Intel® C-612 chipset, Intel Wellsburg Platform</p> <p>Controller Hub (PCH) : 6 x DDR4 DIMMs, 1SPC, 6 x native channels / system</p> <p>Supported speeds: 2133 or 2400MT/s</p> <p>Registered/LRDIMM ECC</p> <p>Memory Populated: 32GB</p> <p>Storage: 256GB Solid State Drive (SSD), 2 TB</p> <p>RAID: Intel RAID supporting 0/1/5</p> <p>Expansion: 1x PCIe 3.0 x16 or 1x PCIe Gen3 x 20 (x16 or x4)</p> <p>Ethernet: 2x Intel® i210 (Springville 1GbE) Controllers</p> <p>External IO: 2x USB 3.0</p> <p>Manageability: Pilot 3 BMC</p> | Dr. K. Hima Bindu     |
| 6. | Graphics card (quantity:2) | GPU GeForce RTX 2080 Ti   | <p>Memory Size: 11GB GDDR6</p> <p>Memory Bus: 352-bit</p> <p>Output: HDMI/3 X Display Porrt/USB Type-c</p> <p>Supports: Windows 10(April 2018 Update or later)/7 (64 bit)</p> <p>Engine boost clock: 1665 MHz</p> <p>Memory clock: 14.0 Gbps</p> <p>PCI Express 3.0</p> <p>Software compatibility - Game Ready Drivers, Microsoft DirectX 12 API, Vulkan API , OpenGL 4.5, Microsoft Windows 10/7 x64</p> <p>Power requirements: 650-watt power supply recommended  295-watt max power consumption</p>                                                                                                                                                          | Dr. Nagesh Bhattu S   |
| 7. | Cluster of 3 Rack Servers  | DELL EMC PowerEdge R440 - | <p>Processor Up to two 2nd Generation Intel® Xeon® Scalable processors, up to 24 cores per processor</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Dr. Srilatha Chebrolu |

|    |             |                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                       |
|----|-------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
|    |             | Quantity 2                                                   | <p>Memory 16 DDR4 DIMM slots, Supports RDIMM /LRDIMM, speeds up to 2666MT/s, 1TB max1</p> <p>Storage controllers</p> <p>Internal Controllers: PERC H330, H730P, H740P, HBA330</p> <p>External Controllers: H840, 12 Gbps SAS</p> <p>HBA Software RAID:S140</p> <p>Internal Boot: Boot Optimized Storage Subsystem (BOSS):HWRAID 2 x M.2 SSDs 240GB, 480GB Internal Dual SD Module2</p> <p>Drive bays Front drive bays: Up to 10 x 2.5” SAS/SATA (HDD/SSD) with up to 4 NVMe SSD max 76.8TB or up to 4 x 3.5 SAS/SATA HDD max 64TB Optional DVD-ROM, DVD+RW</p> <p>Power supplies: Bronze 450W (Cabled PSU), Platinum 550W (Hot plug PSU with full redundancy option)</p> <p>Sizing Form factor: Rack (1U)</p>         |                       |
| 8. | Rack Server | <p>HPE DL380 Gen9 Intel Xeon E5-2667v4</p> <p>Quantity 1</p> | <p>Product Type: Processor Upgrade</p> <p>Product Family: Xeon E5-2600 v4</p> <p>Processor Core: Octa-core (8 Core)</p> <p>Clock Speed: 3.20 GHz</p> <p>Thermal Design Power: 135 W</p> <p>Type HPE SmartMemoryDDR4 Registered (RDIMM) or Load Reduced (LRDIMM) or Persistent Memory (NVDIMM)</p> <p>DIMM slots available</p> <p>24 (12 DIMM slots per processor, 4 channels per processor, 3 DIMMs per channel)</p> <p>Maximum capacity (LRDIMM)</p> <p>3TB (24 x 128GB LRDIMM at 2400 MHz)</p> <p>Maximum capacity (RDIMM)</p> <p>768GB (24 x 32GB RDIMM at 2400 MHz)</p> <p>Maximum capacity (NVDIMM)</p> <p>128GB (16 x 8GB NVDIMM)</p> <p>NVDIMM support only with the E5-2600v4 processors, and RDIMMs only</p> | Dr. Srilatha Chebrolu |

## DEPARTMENT OF ELECTRICAL ENGINEERING

### R&D facilities: List of Major equipment

| S. No | Name of Equipment                                                     | Make/Model                                                                                                                           | Technical Specifications/Features                                                                                                                                                                                                                                                                                                                                                                                                              | Qty | Faculty-In charge    |
|-------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------------------|
| 1.    | <b>ESD work place system</b>                                          | <ul style="list-style-type: none"> <li>Digital Storage Oscilloscope – <b>Make:</b> Tektronix, <b>Model:</b> TBS 1072B-EDU</li> </ul> | <b>Bandwidth:</b> 70Mhz,<br><b>Sampling rate:</b> 1G Sa/s,<br>2 Analog Channels                                                                                                                                                                                                                                                                                                                                                                | 08  | Dr. Kiran Teeparthi  |
|       |                                                                       | <ul style="list-style-type: none"> <li>Function Generator-<br/> <b>Make:</b> Tektronix,<br/> <b>Model:</b> AFG 1022,</li> </ul>      | Dual Channel,<br><b>Frequency:</b> 1μHz – 5Mhz,<br><b>Sampling rate:</b> 125M Sa/s                                                                                                                                                                                                                                                                                                                                                             |     |                      |
|       |                                                                       | <ul style="list-style-type: none"> <li>Regulated Power Supply-<br/> <b>Make:</b> Keithley<br/> <b>Model:</b> 2231A-30-3</li> </ul>   | <b>Number of output channels:</b> 3<br><ul style="list-style-type: none"> <li>Output a total of 195W of power with two 0-30V/0-3A outputs and one 0-5V/0-3A</li> </ul>                                                                                                                                                                                                                                                                         |     |                      |
|       |                                                                       | <ul style="list-style-type: none"> <li>Digital Multimeter<br/> <b>Make:</b> Keithley,<br/> <b>Model:</b> 2110</li> </ul>             | <b>DC voltage:</b> 0.1 V, 1 V, 10 V, 100 V, and 1000 V<br><b>AC voltage:</b> 0.1 V, 1 V, 10 V, 100 V, and 750 V,<br><b>DC current:</b> 10 mA, 100 mA, 1 A, 3 A, and 10 A,<br><b>AC current:</b> 1 A, 3 A, and 10 A<br><ul style="list-style-type: none"> <li>Two and 4-wire resistance: 100 Ω, 1 kΩ, 10 kΩ, 100 kΩ, 1 MΩ, 10 MΩ, and 100 MΩ</li> <li>Capacitance measurement: 1 nF, 10 nF, 100 nF, 1 μF, 10 μF, 100 μF, 1 mF, 10 mF</li> </ul> |     |                      |
| 2.    | <b>Artificial Transmission Line Module</b><br>(400KV, 200 Kilometres) | <b>Make:</b> Power Research & Development Consultants Pvt. Ltd, Bangalore                                                            | <b>Source Panel:</b> <ul style="list-style-type: none"> <li>440/110 V isolated Source Transformer (3 Phase)</li> <li>ELC Breaker: FH204 AC-40V/0.03A</li> <li>3 Phase Non-Directional Over Current Relay: EOC03, 110 V</li> <li>VAF Meter: EL Measure, OM 1300</li> <li>Multi-Function Meter: EL</li> </ul>                                                                                                                                    | 01  | Dr. Sankar Peddapati |

|    |                                                                              |                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |                      |
|----|------------------------------------------------------------------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----------------------|
|    |                                                                              |                                                                           | <p>Measure, EN 8400</p> <ul style="list-style-type: none"> <li>• Nominal Current- 1A</li> <li>• Input &amp; Output Voltage: 440V/20-130V (Continuously Variable)</li> <li>• RS 485 based communication</li> </ul> <p><b>PI Section Transmission Line:</b></p> <ul style="list-style-type: none"> <li>• 400kV Line with 3 Phase Twin Moose Conductor of 200 KM length with taps at 50Km, 100Km, 150Km 200Km</li> <li>• Nominal Voltage/Nominal Current: 110V, 1A</li> </ul> <p><b>Load Station Model:</b></p> <ul style="list-style-type: none"> <li>• 3 Separate Multifunction Meter for Input, Load and Capacitor Bank: EL Measure, EN 8400</li> <li>• 3 Phase Non-Directional Over Current Relay: EOC03, 110 V</li> <li>• RS 485 based communication</li> <li>• Inductive, Resistive Loading with Continuous Variable Control</li> <li>• 6 Step Capacitor Bank with Step Control</li> <li>• Nominal Voltage/Nominal Current: 110V, 1A</li> </ul> |    |                      |
| 3. | <b>Fault studies on AC Network Analyser and Sequence Components Analyser</b> | <b>Make:</b> Power Research & Development Consultants Pvt. Ltd, Bangalore | <p><b>Sequence Components Analyser:</b></p> <ul style="list-style-type: none"> <li>• 440 / 110 V Isolated Source Transformer (3 Phase)</li> <li>• Nominal Current – 1A, Fault Current – 5A (10 Sec), 10A (1.2 Sec)</li> <li>• 3 Phase Non-Directional Over Current Relay- EOC-03F, 110 V</li> <li>• Integrated negative sequence relay-ENS-33</li> <li>• Multifunction Meter: :</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 01 | Dr. Sankar Peddapati |

|    |                                            |                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |    |               |
|----|--------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|---------------|
|    |                                            |                                                                 | EL Measure, EN 8400<br><br><b>Motor- Generator Set:</b> <ul style="list-style-type: none"> <li>• DC Motor:3HP, 220V, 12A, 1500 R.P.M</li> <li>• Alternator: 2KVA,415V,2.8A, 1500 R.P.M</li> </ul> <b>PI Section Transmission Line:</b> <ul style="list-style-type: none"> <li>• 400kV Line with 3 Phase Twin Moose Conductor of 200 KM length with taps at 50Km, 100Km, 150Km 200Km</li> <li>• Nominal Voltage/Nominal Current: 110V, 1A</li> </ul> <b>Fault Simulator:</b> <ul style="list-style-type: none"> <li>• LG / LL / LLG / LLL / LLLG Fault with Phase Selectable LG / LL Fault</li> </ul> |    |               |
| 4. | <b>Digital Storage Oscilloscopes</b>       | <b>Make:</b> Rigol<br><b>Model:</b> DS1204B                     | <b>Bandwidth:</b> 200 MHz,<br><b>Sampling rate:</b> 2 GSa/s, 4 Analog Channels                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 02 | Dr. Jayaram N |
| 5. |                                            | <b>Make:</b> Rigol<br><b>Model:</b> MSO1104Z                    | <b>Bandwidth:</b> 100Mhz,<br><b>Sampling rate:</b> 1 GSa/s, 4 Analog Channels                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 01 | Dr. Jayaram N |
| 6. |                                            | <b>Make:</b> Rigol<br><b>Model:</b> MSO05204                    | <b>Bandwidth:</b> 200Mhz,<br><b>Sampling rate:</b> 8GSa/s, 4 Analog Channels                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 03 | Dr. Jayaram N |
| 7. | <b>Single-Phase Power Quality Analyzer</b> | <b>Make:</b> Fluke<br><b>Model:</b> 125B Industrial scope meter | <ul style="list-style-type: none"> <li>• Dual-input digital oscilloscope and Multimeter</li> <li>• 40 MHz oscilloscope bandwidth</li> <li>• Power measurements (W, VA, VAR, PF, DPF, Hz),</li> <li>• Voltage, current and power harmonics</li> </ul>                                                                                                                                                                                                                                                                                                                                                 | 01 | Dr. Jayaram N |

|     |                                           |                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |                                      |
|-----|-------------------------------------------|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|--------------------------------------|
| 8.  | <b>Three-Phase Power Quality Analyzer</b> | <b>Make:</b> Kusam<br><b>Model:</b> KM-2200  | <ul style="list-style-type: none"> <li>• Easy operation to capture the power quality problem.</li> <li>• 8G Memory Card to Make the long-term measuring data storage.</li> <li>• Volts / Amps. / Hertz, Dips &amp; Swell, Power / Energy, Unbalance, Monitor, Scope, Harmonic &amp; Interruption functions</li> <li>• Current Probes 50A – 4 Qty</li> <li>• Current Probes 100A – 4Qty</li> </ul>                                  | 02 | Dr. Jayaram N                        |
| 9.  | <b>dSPACE</b>                             | <b>Make:</b> dSPACE<br><b>Model:</b> 1104    | <b>Hardware:</b> <ul style="list-style-type: none"> <li>• DS1104 R&amp;D Controller Board PCIe Version, MPC8240 processor with PPC 603e core, CPU clock: 250 MHz and 32 MB SDRAM</li> <li>• CLP 1104-connector &amp; LED panel for DS 1104</li> </ul> <b>Software:</b> <ul style="list-style-type: none"> <li>• CDP1104 control development software package with USB dongle</li> <li>• Microtec PowerPC cross compiler</li> </ul> | 02 | Dr. Jayaram N & Dr. Tejavathu Ramesh |
| 10. | <b>Visual IR Thermometer</b>              | <b>Make:</b> Fluke<br><b>Model:</b> VT02A    | <b>Temperature measurement range:</b><br>-10 °C to +250 °C (14 °F to 482 °F)<br><b>Temperature measurement accuracy:</b> ± 2 °C or ± 2 %<br><b>Image blending:</b><br>0%, 25%, 50%, 75%, 100%                                                                                                                                                                                                                                      | 01 | Dr. Jayaram N                        |
| 11. | <b>Programmable DC Electronic Load</b>    | <b>Make:</b> Tenma<br><b>Model:</b> 72-13210 | <b>Voltage Measurement:</b> <ul style="list-style-type: none"> <li>• Range: 0-18V / 0-120V</li> <li>• Resolution: 0.1mV/1mV</li> <li>• Accuracy: ±(0.025% + 0.025% FS)</li> </ul> <b>Current Measurement:</b>                                                                                                                                                                                                                      | 01 | Dr. Jayaram N                        |

|     |                                 |                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |    |                             |
|-----|---------------------------------|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----------------------------|
|     |                                 |                                                    | <ul style="list-style-type: none"> <li>• Range: 0-3A / 0-30A</li> <li>• Resolution: 0.1mA / 1mA</li> <li>• Accuracy: <math>\pm(0.05\% + 0.05\% \text{ FS})</math></li> </ul> <b>Power Measurement:</b> <ul style="list-style-type: none"> <li>• Range: 300W</li> <li>• Resolution: 0.01W</li> <li>• Accuracy: <math>\pm(0.1\% + 0.1\% \text{ FS})</math></li> </ul> <b>Features:</b> <ul style="list-style-type: none"> <li>• 4 working modes: CV/CC/CR/CW</li> <li>• Battery test, automatic test, OPP test, OCP test functions</li> </ul> |    |                             |
| 12. | <b>AC/DC Current Probe</b>      | <b>Make:</b> Tektronix<br><b>Model:</b> A622       | <b>Input:</b><br>0 to 70A RMS/100A Peak AC or DC<br><b>Frequency:</b><br>Dc to 100kHz<br><b>Output:</b><br>10 mV/A,100 mV/A                                                                                                                                                                                                                                                                                                                                                                                                                 | 01 | Dr. Jayaram N               |
| 13. | <b>Solar PV Emulator System</b> | <b>Make:</b> ECOSENSE<br><b>Model:</b> SPVE001     | <b>Output Ratings:</b><br>Output Voltage: 0-50 V DC/Channel<br>Output Current: 0-20 A DC/ Channel<br>Output Power: 0-1000 W / Channel<br><b>Line/Load Regulation:</b><br>Voltage: +/- 0.1%<br>Current: +/- 0.1%<br><b>Voltage/ Current Measurement Accuracy:</b><br>0.1%                                                                                                                                                                                                                                                                    | 01 | Dr. Jayaram N               |
| 14. | <b>Tower - Work Station</b>     | <b>Make:</b> Dell<br><b>Model:</b> Precision 5820T | <b>Processor:</b> Intel Xeon W-2265 CPU 3.50GHz<br><b>Memory:</b> 32 GB<br><b>Storage:</b> 500 GB HDD                                                                                                                                                                                                                                                                                                                                                                                                                                       | 01 | Dr. Sri Phani Krishna Karri |

**DEPARTMENT OF ELECTRONICS AND COMMUNICATION  
ENGINEERING**

**R&D facilities: List of Major equipment**

| <b>S.NO.</b> | <b>Name of the equipment /Item</b>            |
|--------------|-----------------------------------------------|
| 1            | Klystron Based Test Bench (X Band)            |
| 2            | Gunn Based Test Bench                         |
| 3            | Vector Network Analyzer                       |
| 4            | Spectrum Analyzer with Tracking Generator     |
| 5            | Antenna Radiation Pattern Measurement Set Up  |
| 9            | Power Meter                                   |
| 10           | Noise Source                                  |
| 11           | Experimental Block of OTDR                    |
| 12           | Mode Observation / Characteristics Laboratory |

## DEPARTMENT OF MECHANICAL ENGINEERING

### R&D facilities: List of Major equipment

| S.NO. | Name of the equipment /Item                                                              |
|-------|------------------------------------------------------------------------------------------|
| 1     | CNC LATHE -3Qnty                                                                         |
| 2     | CNC MILL -3-Qnty                                                                         |
| 3     | CONVENTIONAL LATHE -4 Qnty                                                               |
| 4     | UNIVERSAL MILL 2-Qnty                                                                    |
| 5     | MOTORISED PLANER M/C                                                                     |
| 6     | SHAPING MACHINE                                                                          |
| 7     | RADIAL DRILL MACHINE                                                                     |
| 8     | ROBOTIC ARM                                                                              |
| 9     | ELECTRO PNEUMATIC T.K WITH PLC 2-Qnty.                                                   |
| 10    | ELECTRO HYDRAULIC T.K WITH PLC 2-Qnty.                                                   |
| 11    | PLC HMI T.K 2-Qnty.                                                                      |
| 12    | FANUC SIMULATOR 3-Qnty.                                                                  |
| 13    | SEIMENS SIMULATOR (3L+3M) 6-Qnty.                                                        |
| 14    | METHYLENE BLUE CLAY TESTER                                                               |
| 15    | UNIVERSAL STRENGTH MACHINE [hydraulic]                                                   |
| 16    | MOULD HARDNESS TESTER ('B' SCALE)                                                        |
| 17    | Compression Strength testing                                                             |
| 18    | Shear Strength testing                                                                   |
| 19    | MUFFLE WITH INDUCTION FURNACE                                                            |
| 20    | SAND SIEVER                                                                              |
| 21    | SAND RAMMER                                                                              |
| 22    | PERMEABILITY TESTER                                                                      |
| 23    | Arc Welding Machine with necessary accessories                                           |
| 24    | TIG Welding Machine with necessary accessories                                           |
| 25    | MIG Welding Machine with necessary accessories                                           |
| 26    | Submerged arc welding machine with necessary accessories                                 |
| 27    | Plasma arc welding machine with necessary accessories                                    |
| 28    | Spot welding machine with necessary accessories                                          |
| 29    | Automatic movement of welding torch: linear welding fixture                              |
| 30    | Bottom Pouring Stir Casting Furnace                                                      |
| 31    | Pin on Disc Wear Monitoring Apparatus                                                    |
| 32    | Low Speed Wind Tunnel                                                                    |
| 33    | Computational Facility (Work Station) - IC Engines and Battery Thermal Management System |

# DEPARTMENT OF METALLURGICAL AND MATERIALS

## ENGINEERING

### R&D facilities: List of Major equipment

| S.No | Name of the equipment                 |
|------|---------------------------------------|
| 1    | Electronic Balanace(0.0001g)          |
| 2    | Density Measurment Kit                |
| 3    | Doube Disc Polisher                   |
| 4    | Belt Grinder                          |
| 5    | Muffle Furnace(1200°C)                |
| 6    | Electrolytic Cells                    |
| 7    | Poldi Hardness Tester                 |
| 8    | Optical Microscope (Inverted)         |
| 9    | Stereo Zoom Microscope                |
| 10   | Optical Microscope(Upright)           |
| 11   | Distilled Water Equipment Set Up      |
| 12   | Hot Plate (100°C)                     |
| 13   | Magnetic Stirrer with Hot Plate       |
| 14   | pH Meter                              |
| 15   | Digital Stop Watches                  |
| 16   | Laboratory Hot Oven (100°C)           |
| 17   | Ultra sonic Bath                      |
| 18   | Electro Chemical corrosion Analysar   |
| 19   | Tubular Furnace                       |
| 20   | Electrolytic Cell (potentiostat)      |
| 21   | Planetary Ball Mill (to be installed) |

## SCHOOL OF SCIENCE

### R&D facilities: List of Major equipment

| S.NO. | Name of the equipment /Item                                                                                         | Faculty-in-charge      |
|-------|---------------------------------------------------------------------------------------------------------------------|------------------------|
| 1     | Vacuum sealing unit<br>Pelletizer                                                                                   | Dr. Tapas Paramanik    |
| 2     | High temperature Furnace<br>Ball Milling setup<br>Abbe refractometer<br>UV Cabinet<br>Air Oven                      | Dr. R. Arun Kumar      |
| 3     | Workstation,<br>High temperature tubular furnace,<br>Current source meter,<br>Temperature controller, Nanovoltmeter | Dr J Krishnamurthy     |
| 4     | Dell Precision 5820 Tower - Work Station                                                                            | Dr. Sudarshan Dhua     |
| 5     | Dell Precision 5820 Tower - Work Station                                                                            | Dr. Kurmayya Tamminana |

**SCHOOL OF SCIENCES (CHEMISTRY)****R&D facilities: List of Major equipment**

| <b>S. No.</b> | <b>Equipment Name</b>                                                                                        |
|---------------|--------------------------------------------------------------------------------------------------------------|
| 1             | Agilent Cary 60 UV-Visible spectrophotometer                                                                 |
| 2             | Heidolph LCD with Colour Graphic Display Rotary Evaporator, Capacity 5 liter<br>Model : Hei-VAP Core HL/G3   |
| 3             | Equitron Movable with wheels Recirculating chiller, Capacity 15 lpm                                          |
| 4             | GeNei Dual Wavelength Hand held Transilluminators UV Lamp (254 and 365 nm)                                   |
| 5             | iGene Labserve Die cast Aluminium Electronically controlled digital Magnetic Stirrer with temperature sensor |
| 6             | Ocean Life Science Corporation 3000 Watt with 200 Litres Hot Air Oven                                        |
| 7             | Ocean Life Science Corporation 2500 Watt 252 Litres Hot Air Oven                                             |
| 8             | QUASMO UV Cabinet for UV Cabinet for TLC spot visualization (254 and 365 nm)                                 |
| 9             | REMI Stainless Steel Microprocessor Controlled Magnetic Stirrers (Quantity = 3)                              |
| 10            | Athena Technology Cast Iron Vacuum Pump, Air Tank Capacity 45 liter                                          |
| 11            | Digital Analytical Balance (Sartorius)                                                                       |
| 12            | Systronics Microprocessor Based Manual pH Meter                                                              |
| 13            | Systronics Manual Digital Conductivity Meter                                                                 |
| 14            | INSIF INDIA Colorimeter for laboratory, Aluminium with rolled rim Model                                      |
| 15            | Whirlpool 265 L Frost Free Double Door Refrigerator                                                          |
| 16            | Whirlpool 300 L Frost Free Double Door Refrigerator                                                          |
| 17            | Labman Scientific Sonicator System (LMUC-6)                                                                  |
| 18            | Digital Potentiometer (BTC Instruments, Model: BTC-369)                                                      |